

Install Prometheus and Grafana on a Linux EC2 machine, connect Prometheus to Grafana, and create a dashboard to view metrics.

EC2

Instances

Dashboard

AWS Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Images

Instances (1/2)

Save filter sets

Running

Find instance by attribute or tag (case-sensitive)

Instance state = running

Clear filters

Last updated 1 minute ago

Connect

Instance state

Actions

Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs
☐	Monitoring_task	i-073ae66a91dc8c097	Running	t3.micro	Initializing	ap-south-1b	ec2-13-232-255-251.ap...	13.232.255.251	-	-
☒	Worker_node	i-0974ed90e6d19926c	Running	t3.micro	Initializing	ap-south-1b	ec2-5-110-168-22.ap-s...	5.110.168.22	-	-

```
[root@ip-172-31-1-118 ~]# cd /opt
[root@ip-172-31-1-118 opt]# sudo wget https://github.com/prometheus/prometheus/releases/download/v3.4.2/prometheus-3.4.2.linux-amd64.tar.gz
--2026-06-11 11:40:09-- https://github.com/prometheus/prometheus/releases/download/v3.4.2/prometheus-3.4.2.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/30878435-360f-4223-b20b-b41d4549ce67?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-06-11T12A30M3A2Z&srsrcc=attachment%3Bfilename%3Dprometheus-3.4.2.linux-amd64.tar.gz&rscrt=application%2Foctet-stream&skoid=96c2d410-5711-018-11-09&sig=aeddb-aad2ab7ab0k8t2ic:39a6g654-9y7b-w7e9-b12b-9s1bs89bg6dde&st=2026-06-11T11A3A29Y3A2GZ&skes=2026-06-11T11A3A30N3A2SZA&skv=2018-11-09&sig=00HCKV570JazLkq75FwHBm3j9TwOBf4VysEzFFOLUPJ3D8jgtveYJeXeoAIQJKVlQIJCjbHgCioIJiIUz1NiNj9.eyp3CMIOiJnaRwodIUY29tiIYYXvkIjoicmVsZWZfZS1h3NdHMuz2LuHGvMcnlvbNrLbnQuY29tiIwiazV5jiIwaZVSMSim1w4Cc16MTc4MTE4MTHYxMcWlbJaiJoXnzgxMt4QMDELcj3wXYRoIjoicmVsZWZfZWFzc2VhCGJvZHJhdGVlbSibG9glzMvcmdudU2luZG93cy5uZXiQifQ.zoETKm djnBiVnklUbVBDeISbwKTTGjYZhneQG-AZFwRresponse-content-disposition=attachment%3B%20filename%3Dprometheus-3.4.2.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-06-11 11:40:10-- https://release-assets.githubusercontent.com/github-production-release-asset/6838921/30878435-360f-4223-b20b-b41d4549ce67?sp=r&sv=2018-11-09&sr=b&r=httpshttps=2026-06-11T12A30M3A2Z&srsrcc=attachment%3Bfilename%3Dprometheus-3.4.2.linux-amd64.tar.gz&rscrt=application%2Foctet-stream&skoid=96c2d410-5711-43al-aedd-ab2ab7ab0k8t2ic:39a6g654-9y7b-w7e9-b12b-9s1bs89bg6dde&st=2026-06-11T11A3A29Y3A2GZ&skes=2026-06-11T12A30N3A2SZA&skv=2018-11-09&sig=00HCKV570JazLkq75FwHBm3j9TwOBf4VysEzFFOLUPJ3D8jgtveYJeXeoAIQJKVlQIJCjbHgCioIJiIUz1NiNj9.eyp3CMIOiJnaRwodIUY29tiIYYXvkIjoicmVsZWZfZS1h3NdHMuz2LuHGvMcnlvbNrLbnQuY29tiIwiazV5jiIwaZVSMSim1w4Cc16MTc4MTE4MTHYxMcWlbJaiJoXnzgxMt4QMDELcj3wXYRoIjoicmVsZWZfZWFzc2VhCGJvZHJhdGVlbSibG9glzMvcmdudU2luZG93cy5uZXiQifQ.zoETKmdjnBiVnklUbVBDeISbwKTTGjYZhneQG-AZFwRresponse-content-disposition=attachment%3B%20filename%3Dprometheus-3.4.2.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.110.133, 185.199.111.133, 185.199.109.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.110.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 117282805 (112M) [application/octet-stream]
Saving to: 'prometheus-3.4.2.linux-amd64.tar.gz'

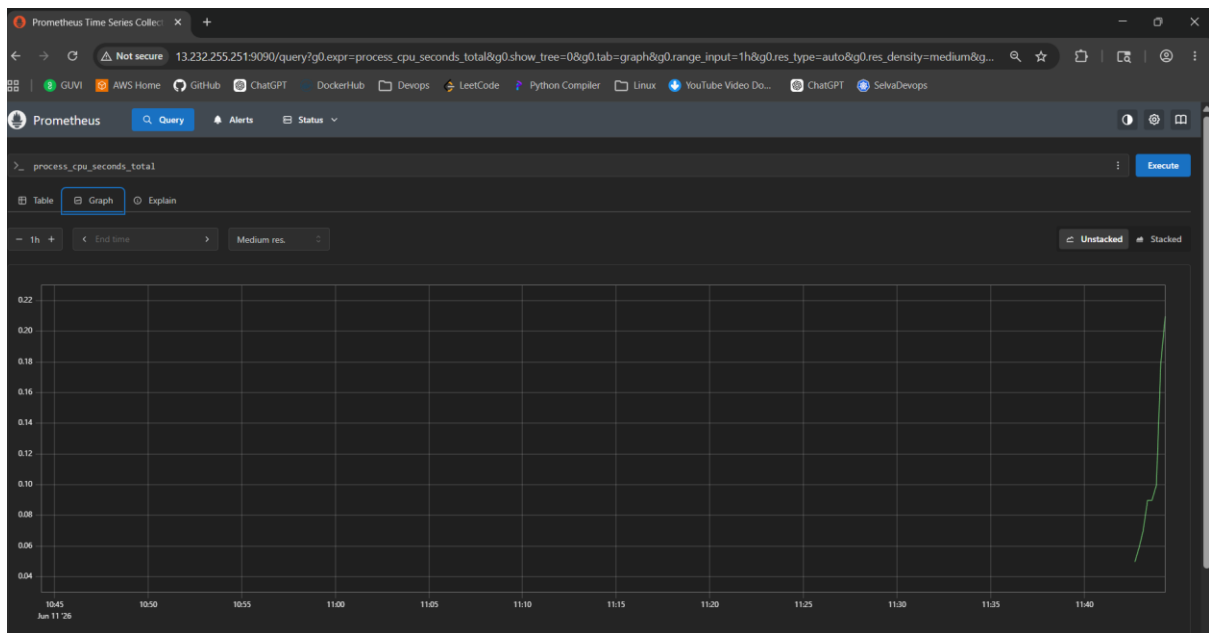
prometheus-3.4.2.linux-amd64. 100%[=====] 111.85M 27.1MB/s in 4.1s

2026-06-11 11:40:14 (27.1 MB/s) - 'prometheus-3.4.2.linux-amd64.tar.gz' saved [117282805/117282805]

[root@ip-172-31-1-118 opt]#
```

```
total 288M
-rwxr-xr-x. 1 root 118 148M Jun 26 2025 prometheus
-rwxr-xr-x. 1 root 118 140M Jun 26 2025 promtool
-rw-r--r--. 1 root 118 1.1M Jun 26 2025 prometheus.yml
-rw-r--r--. 1 root 118 3.7M Jun 26 2025 NOTICE
-rw-r--r--. 1 root 118 12M Jun 26 2025 LICENSE
[root@ip-172-31-1-118 prometheus]# ./prometheus \
--config.file=prometheus.yml
time=2026-06-11T11:42:08.313Z level=INFO source=main.go:674 msg="No time or size retention was set so using the default time retention" duration=15d
time=2026-06-11T11:42:08.313Z level=INFO source=main.go:725 msg="Starting Prometheus Server" mode=server version="(version=3.4.2, branch=HEAD, revision=b392
caf256d7ed3698989d2946ca86274e5557d36)"
time=2026-06-11T11:42:08.313Z level=INFO source=main.go:738 msg="operational information" build_context="(go=go1.24.4, platform=linux/amd64, user=root@f841c
a3a66eb, date=20250626-20:51:48, tags=netgo,builtinassets,stringlabels)" host_details="(Linux 6.1.174-217.345.amzn2023.x86_64 #1 SMP PREEMPT_DYNAMIC Mon Jun
16 16:38:11 UTC 2026 x86_64 ip-172-31-1-118.ap-south-1.compute.internal (none))" fd_limits="(soft=65535, hard=65535)" vm_limits="(soft=unlimited, hard=unli
mited)"
time=2026-06-11T11:42:08.315Z level=INFO source=main.go:806 msg="Leaving GOMAXPROCS=2: CPU quota undefined" component=automaxprocs
time=2026-06-11T11:42:08.318Z level=INFO source=web.go:656 msg="Start listening for connections" component=web address=0.0.0.0:9090
time=2026-06-11T11:42:08.319Z level=INFO source=main.go:1266 msg="Starting TSDB ..."
time=2026-06-11T11:42:08.320Z level=INFO source=tsls_config.go:347 msg="Listening on" component=web address=[::]:9090
time=2026-06-11T11:42:08.321Z level=INFO source=tsls_config.go:350 msg="TLS is disabled." component=web http2=false address=[::]:9090
time=2026-06-11T11:42:08.322Z level=INFO source=head.go:657 msg="Replaying on-disk memory mappable chunks if any" component=tsdb
time=2026-06-11T11:42:08.322Z level=INFO source=head.go:744 msg="On-disk memory mappable chunks replay completed" component=tsdb duration=2.161us
time=2026-06-11T11:42:08.323Z level=INFO source=head.go:752 msg="Replaying WAL, this may take a while" component=tsdb
time=2026-06-11T11:42:08.323Z level=INFO source=head.go:825 msg="WAL segment loaded" component=tsdb segment=0 maxSegment=0 duration=167.46us
time=2026-06-11T11:42:08.323Z level=INFO source=head.go:862 msg="WAL replay completed" component=tsdb checkpoint_replay_duration=56.408us wal_replay_duratio
n=228.277us wbl_replay_duration=132ns chunk_snapshot_load_duration=8s mmap_chunk_replay_durations=2.161us total_replay_duration=438.956us
time=2026-06-11T11:42:08.329Z level=INFO source=main.go:1287 msg="filesystem information" fs_type=XFS_SUPER_MAGIC
time=2026-06-11T11:42:08.329Z level=INFO source=main.go:1290 msg="TSDB started"
time=2026-06-11T11:42:08.329Z level=INFO source=main.go:1475 msg="Loading configuration file" filename=prometheus.yml
time=2026-06-11T11:42:08.333Z level=INFO source=main.go:1514 msg="updated GOGC" old=180 new=75
time=2026-06-11T11:42:08.333Z level=INFO source=main.go:1524 msg="Completed loading of configuration file" db_storage=956ns remote_storage=1.301us web_handl
er=439ns query_engine=5.798us scrape=3.319765ms scrape_sd=26.259us notify=117.59us notify_sd=26.616us rules=1.393us tracing=4.897us filename=prometheus.yml
totalDuration=4.092779ms
time=2026-06-11T11:42:08.333Z level=INFO source=main.go:1251 msg="Server is ready to receive web requests."
time=2026-06-11T11:42:08.333Z level=INFO source=manager.go:175 msg="Starting rule manager..." component="rule manager"
```

Prometheus working:



Installing Grafana:

```
[root@ip-172-31-1-118 ~]# sudo vi /etc/yum.repos.d/grafana.repo
[root@ip-172-31-1-118 ~]# sudo dnf install grafana -y
Amazon Linux 2023 Kernel Livepatch repository
grafana
grafana
Importing GPG key 0x10458545:
  Userid      : "Grafana Labs <engineering@grafana.com>"
  Fingerprint: B53A E77B AD86 30A6 8304 6085 963F A277 1045 8545
  From        : https://rpm.grafana.com/gpg.key
grafana
Killed
[root@ip-172-31-1-118 ~]# grafana-server -v
-bash: grafana-server: command not found
[root@ip-172-31-1-118 ~]# sudo dnf install grafana -y
Last metadata expiration check: 0:01:08 ago on Thu Jun 11 11:52:24 2026.
Dependencies resolved.
=====
Package                               Architecture      Version           Repository        Size
Installing:
grafana                               x86_64            13.0.2-1         grafana           212 M
=====
106 MB/s | 92 MB      00:00
```

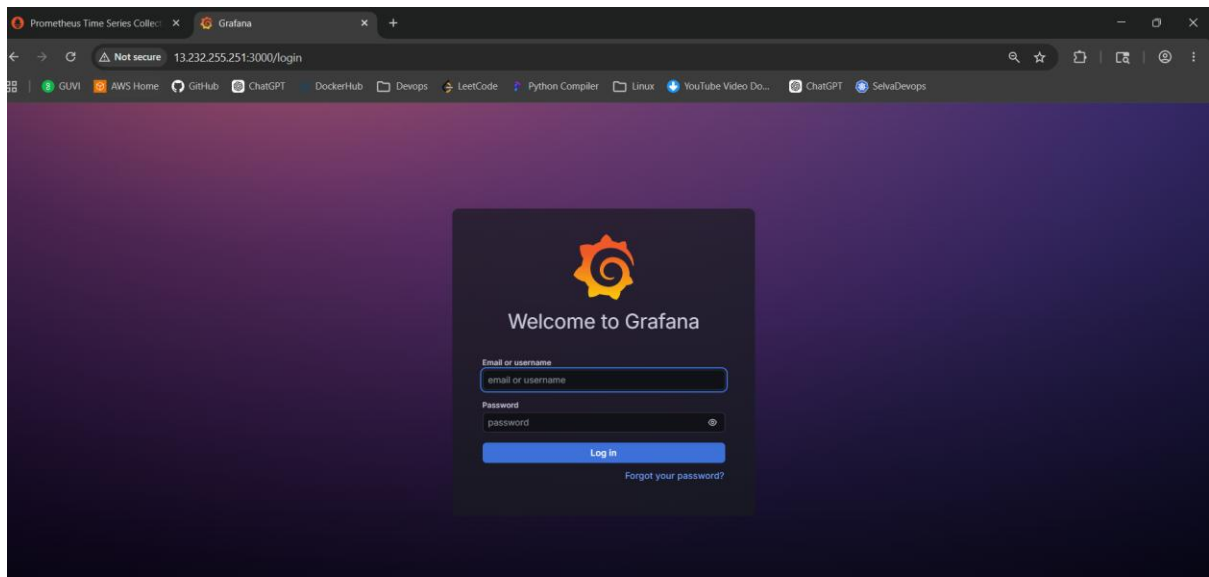
```
Installed:
  grafana-13.0.2-1.x86_64

Complete!
[root@ip-172-31-1-118 ~]# grafana-server -v
Deprecation warning: 'grafana-server' is deprecated and will be removed in a future release. Use the 'grafana server' subcommand instead.
Version 13.0.2 (commit: 3fcdbc5a, branch: release-13.0.2)
[root@ip-172-31-1-118 ~]# |
```

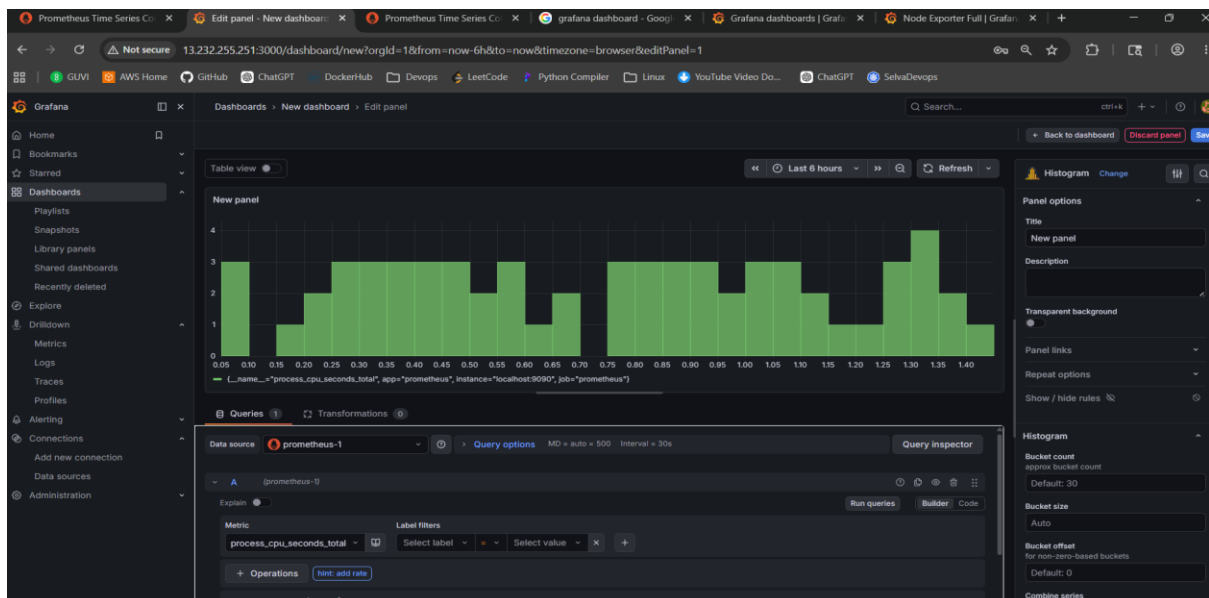
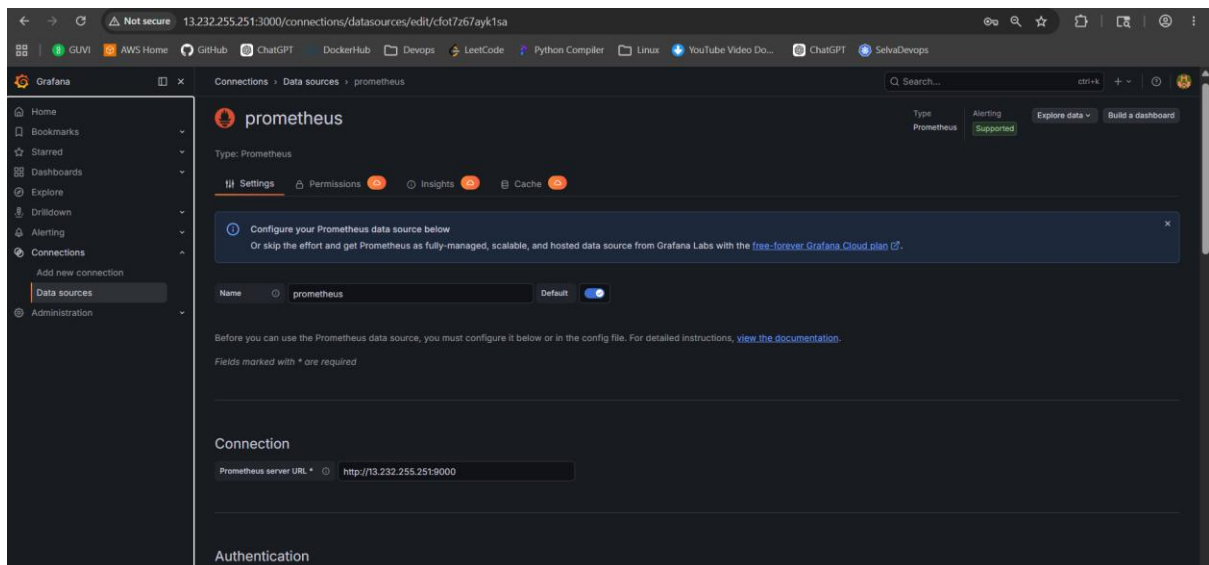
Grafana-server running:

```
Version 13.0.2 (commit: 3fcdbc5a, branch: release-13.0.2)
[root@ip-172-31-1-118 ~]# sudo systemctl daemon-reload
sudo systemctl enable grafana-server
sudo systemctl start grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /usr/lib/systemd/system/grafana-server.service.
[root@ip-172-31-1-118 ~]# systemctl status grafana-server
● grafana-server.service - Grafana instance
   Loaded: loaded (/usr/lib/systemd/system/grafana-server.service; enabled; preset: disabled)
   Active: active (running) since Thu 2026-06-11 11:55:40 UTC; 15s ago
     Docs: http://docs.grafana.org
    Main PID: 2739 (grafana)
      Tasks: 28 (Limit: 1014)
    Memory: 262.2M
       CPU: 4.781s
    CGroup: /system.slice/grafana-server.service
            └─2739 /usr/share/grafana/bin/grafana server --config=/etc/grafana/grafana.ini --pidfile=/var/run/grafana/grafana-server.pid --packaging=rpm c
              └─2746 /usr/share/grafana/data/plugins-bundled/elasticsearch/gpx_grafana_elasticsearch_datasource_linux_amd64

Jun 11 11:55:44 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugin.backgroundinstaller t=2026-06-11T11:55:44.479128795Z level=info ms
Jun 11 11:55:45 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugin.installer t=2026-06-11T11:55:45.376575664Z level=info msg="Install
Jun 11 11:55:45 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=installer.fs t=2026-06-11T11:55:45.470889855Z level=info msg="Downloaded >
Jun 11 11:55:45 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugins.registration t=2026-06-11T11:55:45.520660173Z level=info msg="Plu
Jun 11 11:55:45 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugin.backgroundinstaller t=2026-06-11T11:55:45.52072765Z level=info msg=
Jun 11 11:55:45 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugin.backgroundinstaller t=2026-06-11T11:55:45.520758916Z level=info msg=
Jun 11 11:55:46 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugin.installer t=2026-06-11T11:55:46.275645291Z level=info msg="Install
Jun 11 11:55:46 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=installer.fs t=2026-06-11T11:55:46.369306494Z level=info msg="Downloaded >
Jun 11 11:55:46 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugins.registration t=2026-06-11T11:55:46.417595154Z level=info msg="Plu
Jun 11 11:55:46 ip-172-31-1-118.ap-south-1.compute.internal grafana[2739]: logger=plugin.backgroundinstaller t=2026-06-11T11:55:46.417651527Z level=info ms
lines 1-22/22 (END)
```



Creating data source:



Installing Node exporter in worker node machines:

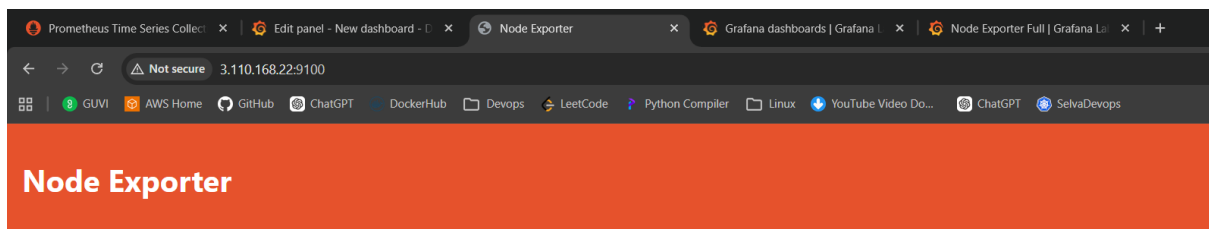
```
[root@ip-172-31-13-61 ~]# cd /opt
[root@ip-172-31-13-61 ~]# sudo wget https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
--2026-06-11 12:16:46-- https://github.com/prometheus/node_exporter/releases/download/v1.9.1/node_exporter-1.9.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/dc8ec09c-2975-42a2-9591-57dd1ffff7b7?sp=r&sv=2018-11-09&sc=b6spr=https%3A%2F%2Fgithub.com%2Fassets%2Fnode_exporter-1.9.1.linux-amd64.tar.gz&rsrcct=application/octet-stream&kold=96c2zd4io-5743ai-aedd-ab1947aab08sktld=398ae654-997b-47e9-b12b-9515b896hdde8ske=2026-06-11112331A16X3ASZ8ske=2026-06-11112331A16X3ASZ8ske=b8skyvz2018-11-09&sig=Psd3
x2FPmSzL-Pad3k2FPmSzLP-FahwGhwFUBuwjz1OAXSDBI-tzt-eJ0EAXOI1JKVIOILChBgcCiOlJIUz1NlN1-jp-Ejc3MlOlJnaXRodWIuY291bi9YXkiOiJotcnVsZWZfZWFZFzc2V0CjovZyHjdGlubGVudD1uc293cyUZnXDIUCGNrcyUzUXoiQg-h6cryEMxCrGhhT133zy2cBBIOverJvhxGLeyVtm2g&rresponse-content-disposition=attachment%3BX20filenamel%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rresponse-content-type=application/octet-stream [following]
--2026-06-11 12:16:47-- https://release-assets.githubusercontent.com/github-production-release-asset/9524057/dc8ec09c-2975-42a2-9591-57dd1ffff7b7?sp=r&sv=2018-11-09&sc=b6spr=https%3A%2F%2Fgithub.com%2Fassets%2Fnode_exporter-1.9.1.linux-amd64.tar.gz&rsrcct=application/octet-stream&kold=96c2zd4io-5743ai-aedd-ab1947aab08sktld=398ae654-997b-47e9-b12b-9515b896hdde8ske=2026-06-11112331A16X3ASZ8ske=2026-06-11112331A16X3ASZ8ske=b8skyvz2018-11-09&sig=Psd3k2FPmSzLP-FahwGhwFUBuwjz1OAXSDBI-tzt-eJ0EAXOI1JKVIOILChBgcCiOlJIUz1NlN1-jp-Ejc3MlOlJnaXRodWIuY291bi9YXkiOiJotcnVsZWZfZWFZFzc2V0CjovZyHjdGlubGVudD1uc293cyUZnXDIUCGNrcyUzUXoiQg-h6cryEMxCrGhhT133zy2cBBIOverJvhxGLeyVtm2g&rresponse-content-disposition=attachment%3BX20filenamel%3Dnode_exporter-1.9.1.linux-amd64.tar.gz&rresponse-content-type=application/octet-stream [following]
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.110.133, 185.199.111.133, 185.199.108.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.110.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 11582410 (11M) [application/octet-stream]
Saving to: ‘node_exporter-1.9.1.linux-amd64.tar.gz’

node_exporter-1.9.1.linux-amd64.tar.gz 100K[=====] 11.04M --.-KB/s in 0.02s

2026-06-11 12:16:47 (455 MB/s) - ‘node_exporter-1.9.1.linux-amd64.tar.gz’ saved [11582410/11582410]
```

```
[root@ip-172-31-13-61 opt]# tar -xzvf node_exporter-1.9.1.linux-amd64.tar.gz
node_exporter-1.9.1.linux-amd64/
node_exporter-1.9.1.linux-amd64/LICENSE
node_exporter-1.9.1.linux-amd64/NOTICE
node_exporter-1.9.1.linux-amd64/node_exporter
[root@ip-172-31-13-61 opt]#
```

Node Exporter working:



Prometheus Node Exporter

Version: (version=1.9.1, branch=HEAD, revision=f2ec547b49af53815038a50265aa2adcd1275959)

- Metrics

Download a detailed report of resource usage (pprof format, from the Go runtime):

- heap usage (memory).
- CPU usage (60 second profile).

To visualize and share profiles you can upload to pprof.me

Adding Node exporter to Prometheus:

```
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
            # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label 'job=<job_name>' to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]
        # The label name is added as a label 'label_name=<label_value>' to any timeseries scraped from this config.
        labels:
          app: "prometheus"
  - job_name: "node_exporter"
    static_configs:
      - targets: ["3.110.168.22:9100"]
~
~
```

Prometheus restarted:

We can able to see the message: Server is ready to receive web request:

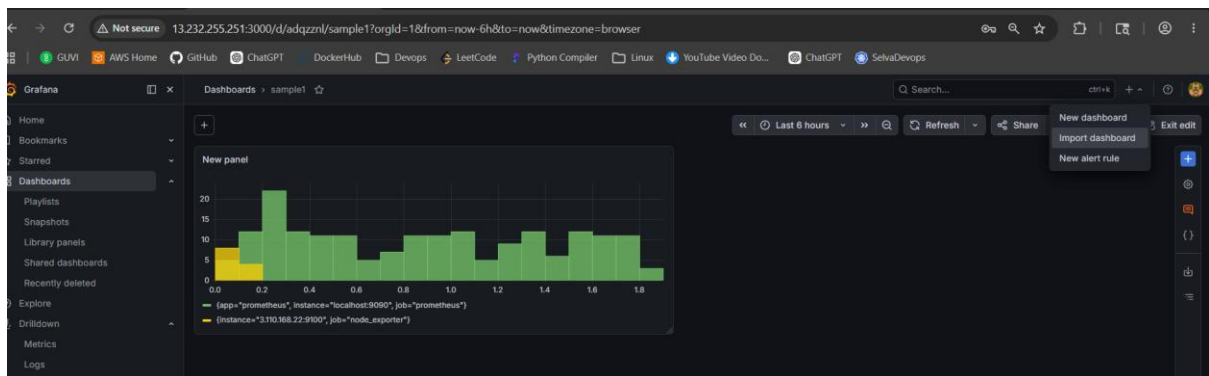
```
time=2026-06-11T12:26:50.783Z level=INFO source=head.go:862 msg="WAL replay completed" component=tsdb checkpoint_replay_duration=48.688µs wal_replay_duration=21.632269ms mbl_replay_duration=3.918µs chunk_snapshot_load_duration=0s mmap_chunk_replay_duration=534.606µs total_replay_duration=22.38313ms
time=2026-06-11T12:26:50.798Z level=INFO source=main.go:1287 msg="filesystem information" fs_type=XFS_SUPER_MAGIC
time=2026-06-11T12:26:50.798Z level=INFO source=main.go:1290 msg="TSDB started"
time=2026-06-11T12:26:50.798Z level=INFO source=main.go:1475 msg="Loading configuration file" filename=prometheus.yml
time=2026-06-11T12:26:50.794Z level=INFO source=main.go:1514 msg="updated GOGC" old=100 new=75
time=2026-06-11T12:26:50.794Z level=INFO source=main.go:1524 msg="Completed loading of configuration file" db_storage=1.100µs remote_storage=1.699µs web_handler=580ns query_engine=7.681µs scrape=3.986264ms scrape_sd=51.729µs notify=143.701µs notify_sd=15.782µs rules=1.611µs tracing=4.179µs filename=prometheus.yml totalDuration=4.651779ms
time=2026-06-11T12:26:50.794Z level=INFO source=main.go:1251 msg="Server is ready to receive web requests."
time=2026-06-11T12:26:50.794Z level=INFO source=manager.go:175 msg="Starting rule manager..." component="rule manager"
```

Prometheus Target list Node Exported added:

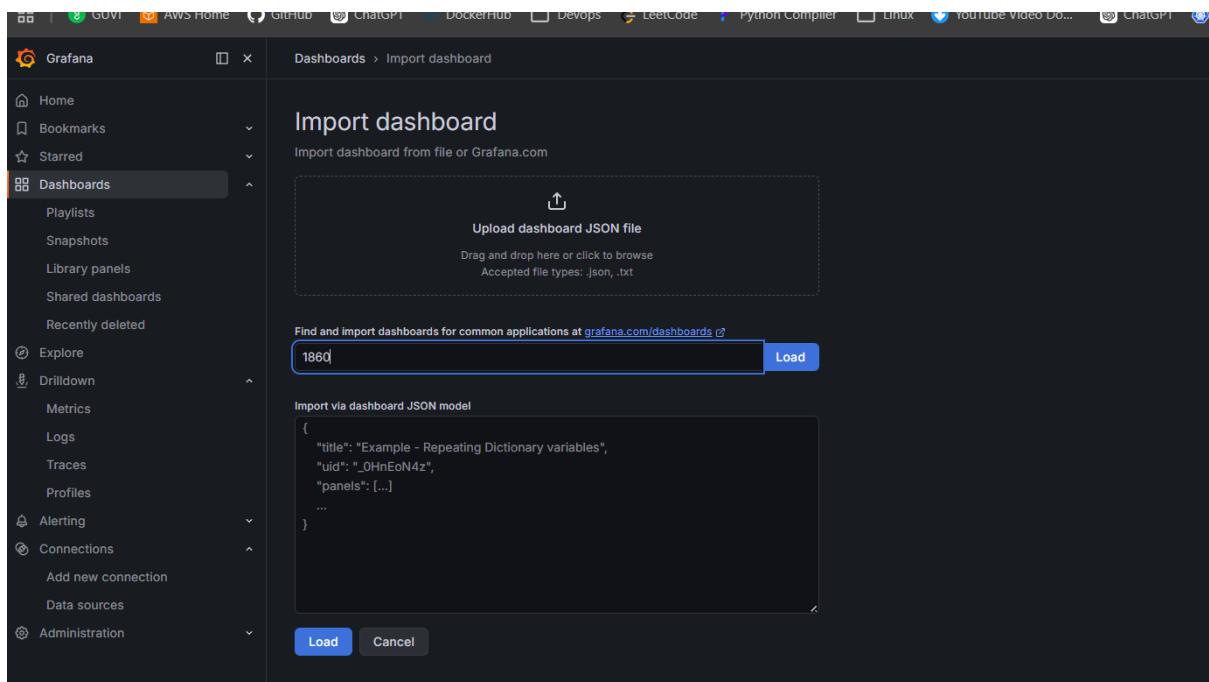
The screenshot shows the Prometheus web interface with the 'Status > Target health' tab selected. The interface displays two target groups: 'node_exporter' and 'prometheus'. Each group has a table of targets with columns for Endpoint, Labels, Last scrape, and State. Both targets are in an 'UP' state.

Target Group	Endpoint	Labels	Last scrape	State
node_exporter	http://3.110.168.22:9100/metrics	instance="3.110.168.22:9100" job="node_exporter"	8.627s ago 67ms	UP
	http://localhost:9090/metrics	app="prometheus" instance="localhost:9090" job="prometheus"	14.896s ago 4ms	UP

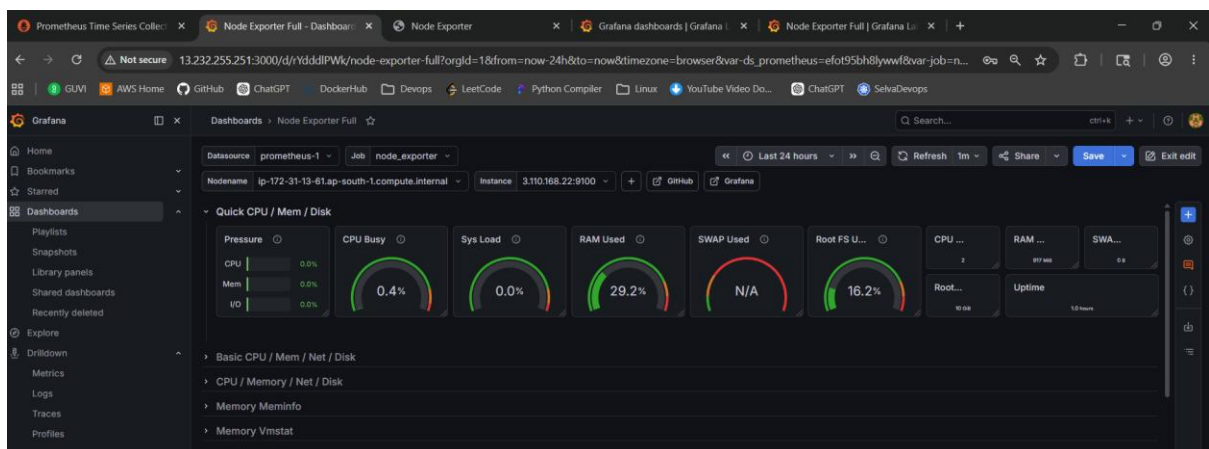
Import New Dashboard:



Adding ID to import Dashboard:



Grafana Dashboard using Node Exporter metrics:



Security group enable:

sg-09521ae51c1784 - launch-wizard-15

Inbound security group rules successfully modified on security group sg-09521ae51c1784 | launch-wizard-15

Details

sg-09521ae51c1784 - launch-wizard-15

Actions

Details

Security group name

launch-wizard-15

Security group ID

sg-09521ae51c1784

Description

launch-wizard-15 created 2026-06-11T11:34:03.555Z

VPC ID

vpc-097050c1a56b9a7f3

Owner

134943601221

Inbound rules count

4 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (4)

Manage tags

Edit inbound rules

Search

	Name	Security group rule ID	IP version	Type	Protocol	Port range	Source	Description
☐	-	sgr-0ad2ac8feb01d8fab	IPv4	Custom TCP	TCP	9100	0.0.0.0/0	-
☐	-	sgr-0ec1874b0cfc8eab8	IPv4	Custom TCP	TCP	3000	0.0.0.0/0	-
☐	-	sgr-057eb3dde313d9c4a	IPv4	SSH	TCP	22	0.0.0.0/0	-
☐	-	sgr-036879acbca2bec9d	IPv4	Custom TCP	TCP	9090	0.0.0.0/0	-